Practical No.: 2

Title: Client-side Form Validation in ReactJS

Aim: Write a ReactJS program to implement client-side form validation.

Theory

Concept of Client-Side Validation in ReactJS

Client-side validation ensures that form inputs meet specific requirements before submitting the data to the server. ReactJS provides controlled components using the `useState()` hook to manage form input values and validate them dynamically.

Types of Validation:

- **Required Field Validation** - Ensures input is not empty.
- **Email Validation** - Checks if the email format is correct.
- **Password Validation** - Ensures minimum length and special character requirements.
- **Confirm Password Validation** - Ensures password and confirm password match.
- **Mobile Number Validation** - Allows only digits and checks length.

Syntax of useState for Form Handling:

```
const [inputValue, setInputValue] = useState("");
const handleChange = (e) => {
    setInputValue(e.target.value);
};
```

- `useState` manages form input values.
- `handleChange` updates state whenever the input field changes.

ReactJS Code

Step 1: Create `index.html`

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>React Form Validation</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
</head>
<body>
    <h2>Practical 2: Client-side Form Validation</h2>
    <div id="root"></div>

    <script type="text/babel" src="formValidation.js"></script>
</body>
</html>
```

Step 2: Create index.js

```
function FormValidation() {
    const [name, setName] = React.useState("");
    const [error, setError] = React.useState("");

    function handleSubmit(e) {
        e.preventDefault();
        if (!name.trim()) {
            setError("Name is required!");
        } else {
            setError("");
            alert("Form submitted successfully!");
        }
    }

    return (
        <form onSubmit={handleSubmit}>
            <label>Enter Name:</label>
            <input type="text" value={name} onChange={(e) => setName(e.target.value)} />
            <p style={{ color: "red" }}>{error}</p>
            <button type="submit">Submit</button>
        </form>
    );
}

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<FormValidation />);
```